(M= Be, Mg, Ca, Sr, Ba, Ra ,
X = F, Cl, Br, l)

Thermal decomposition of $(NH_4)_2BeF_4$ is the best route for the preparation of BeF_2. $BeCl_2$ is conveniently made from the oxide.

$$BeO + C + Cl_2 \xrightarrow{600 - 800K} BeCl_2 + CO$$

Reactivity towards hydrogen:

All the elements except beryllium, combine with hydrogen on heating to form their hydrides with general formula MH_2. BeH_2 can be prepared by the reaction of $BeCl_2$ with $LiAlH_4$.

$$2BeCl_2 + LiAlH_4 \rightarrow 2BeH_2 + LiCl + AlCl_3$$

5.5.4 Uses of alkaline earth metals

Uses of beryllium

1. Because of its low atomic number and very low absorption for X-rays, it is used as radiation windows for X-ray tubes and X-ray detectors.
2. The sample holder in X-ray emission studies usually made of beryllium
3. Since beryllium is transparent to energetic particles, it is used to build the 'beam pipe' in accelerators.
4. Because of its low density and diamagnetic nature, it is used in various detectors.

Uses of magnesium

1. Removal of sulphur from iron and steel.
2. Used as photoengrave plates in printing industry.
3. Magnesium alloys are used in aeroplane and missile construction.
4. Mg ribbon is used in synthesis of Grignard reagent in organic synthesis.
5. It alloys with aluminium to improve its mechanical, fabrication and welding property.
6. As a desiccant .
7. As sacrificial anode in controlling galvanic corrosion.

Uses of calcium

1. As a reducing agent in the metallurgy of uranium, zirconium and thorium.
2. As a deoxidiser, desulphuriser or decarboniser for various ferrous and non-ferrous alloys.
3. In making cement and mortar to be used in construction.
4. As a getter in vacuum tubes.
5. In dehydrating oils
6. In fertilisers, concrete and plaster of paris.

Uses of strontium

1. ^{90}Sr is used in cancer therapy.
2. ^{87}Sr / ^{86}Sr ratios are commonly used

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